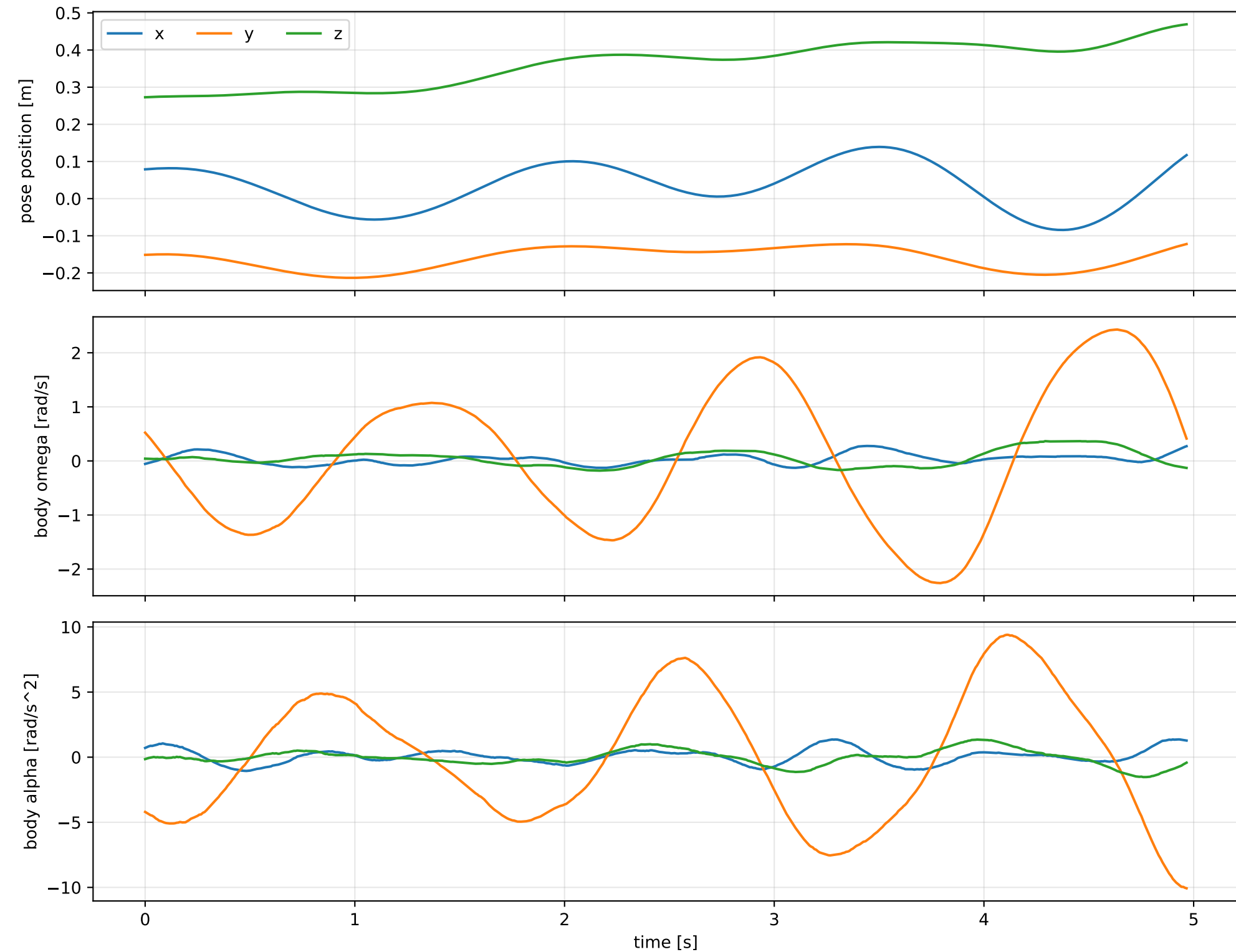
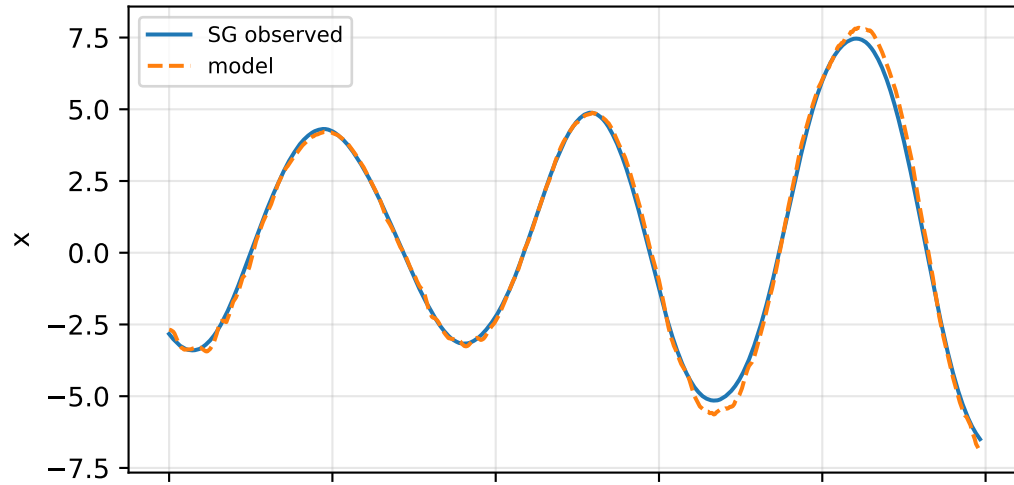


gimbal_measured_linear: SG trajectory and derived motion

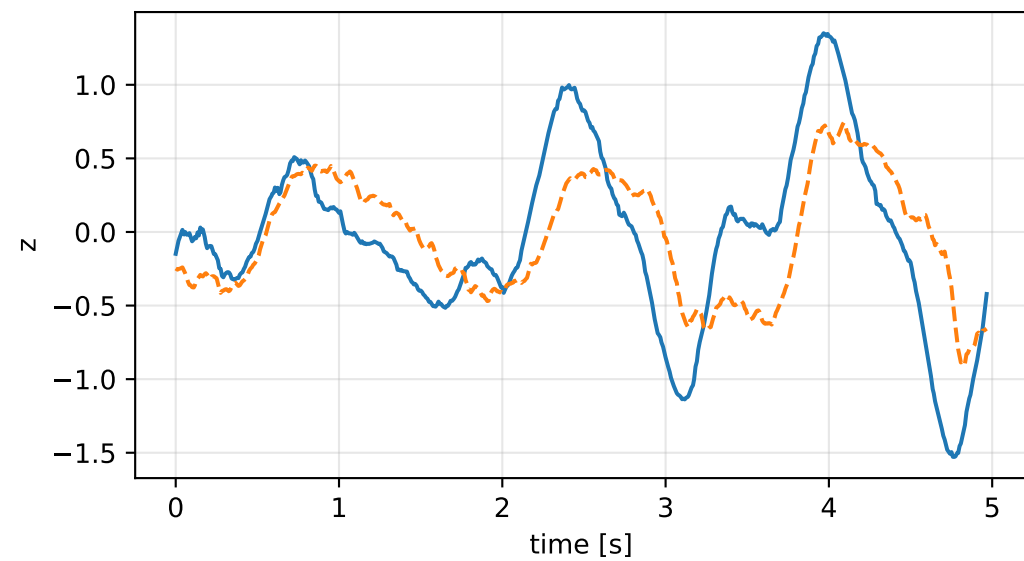
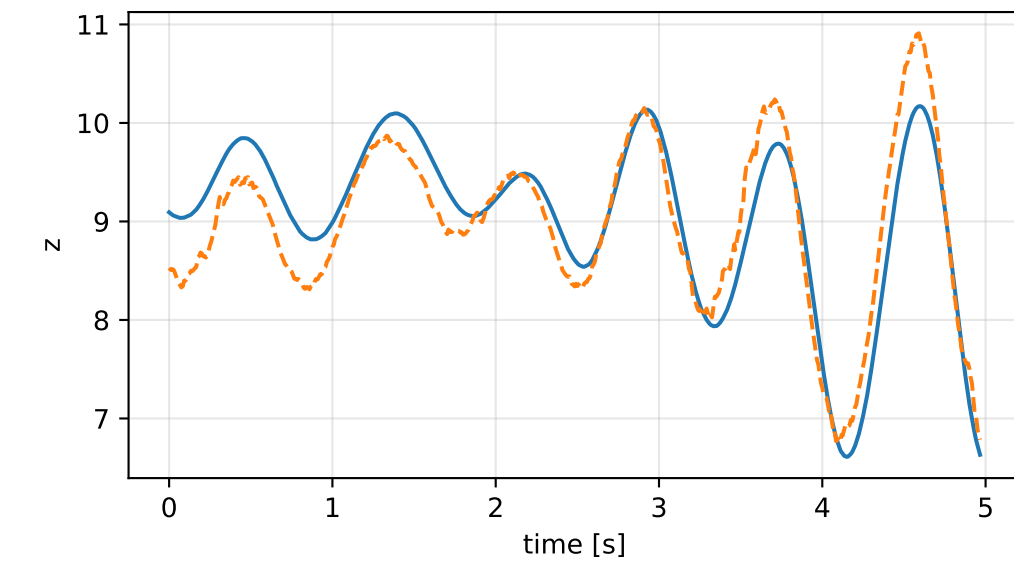
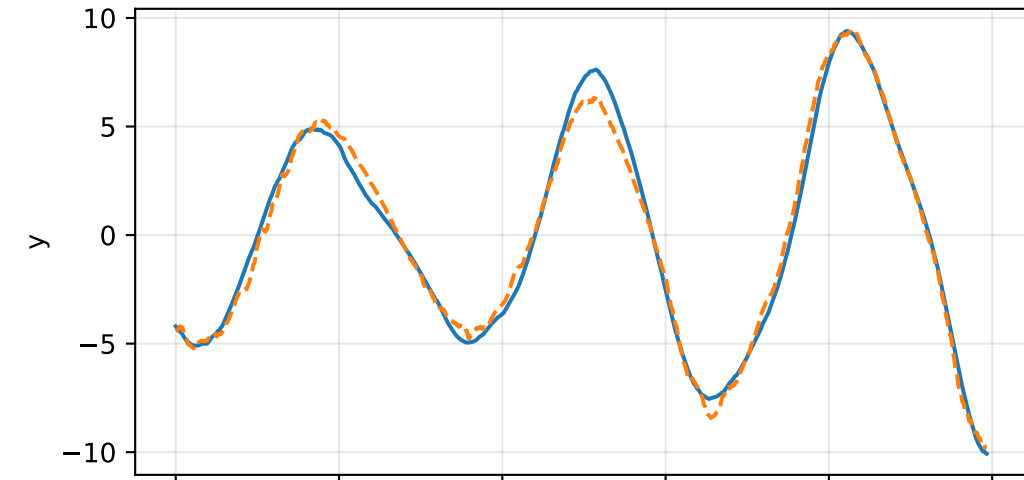
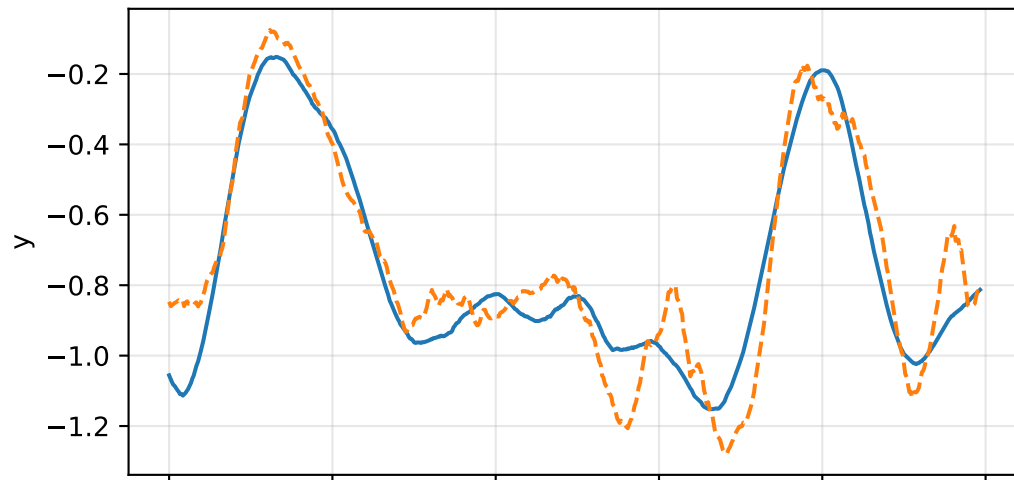
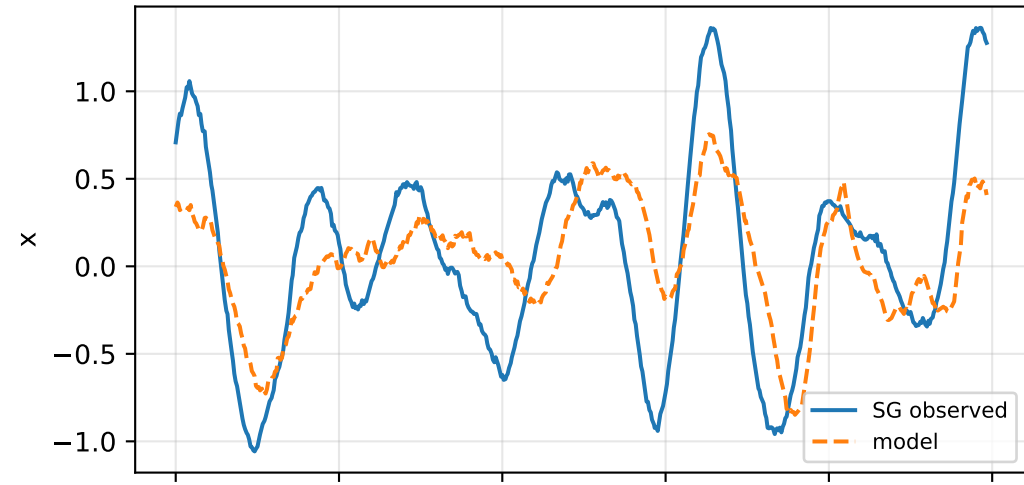


gimbal_measured_linear: acceleration residual objective

specific acceleration [m/s^2]

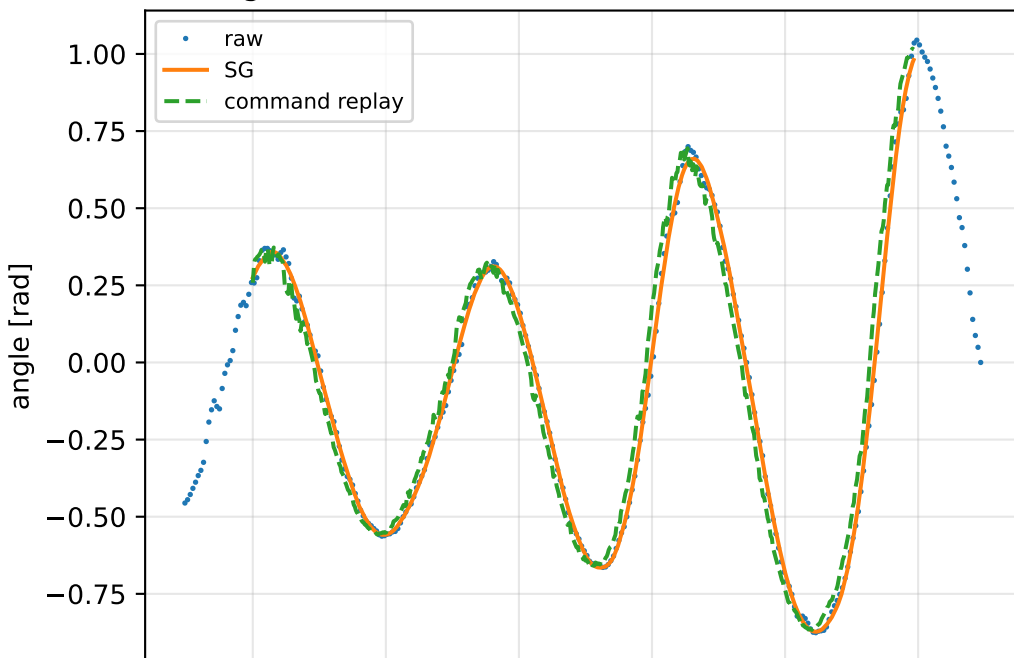


angular acceleration [rad/s^2]

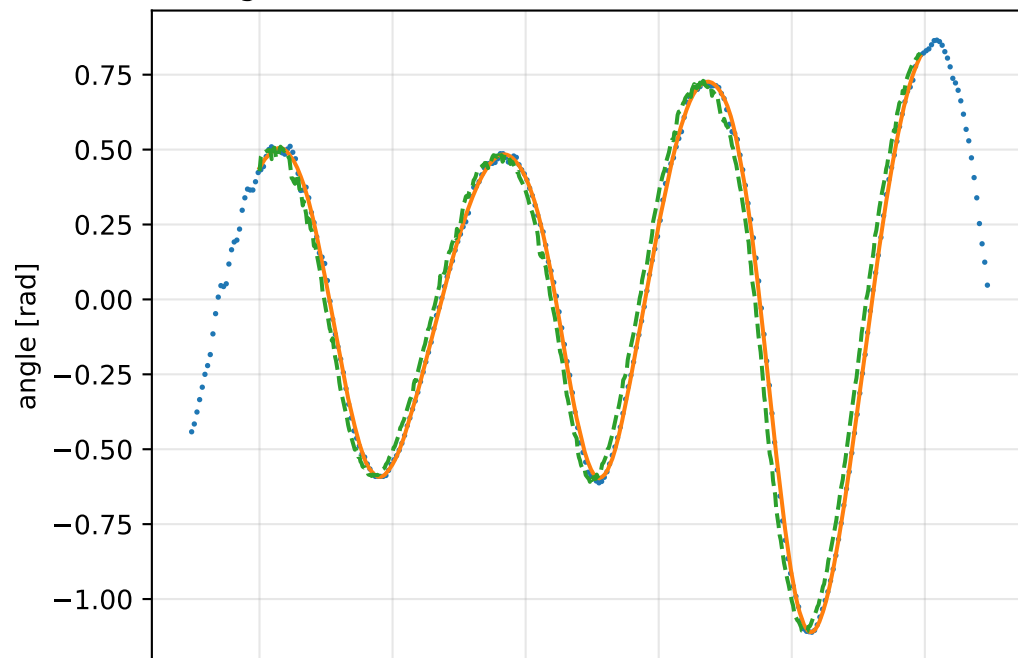


gimbal_measured_linear: gimbal measurement audit (objective=measured_linear)

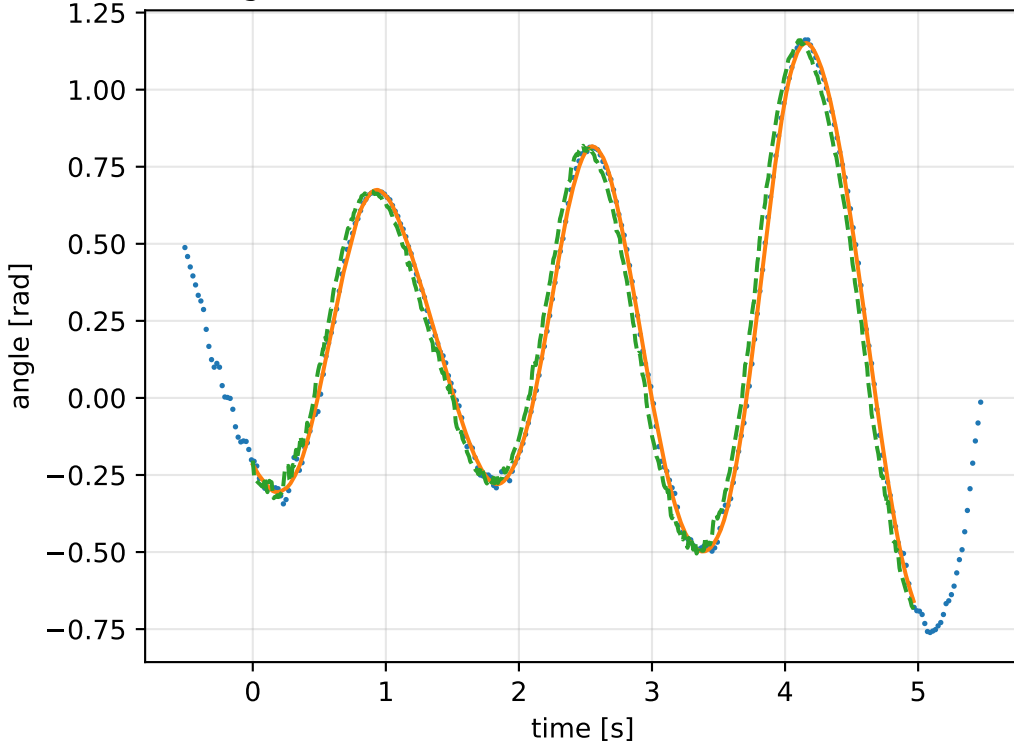
gimbal 1: RMSE=0.0759, max=0.2095 rad



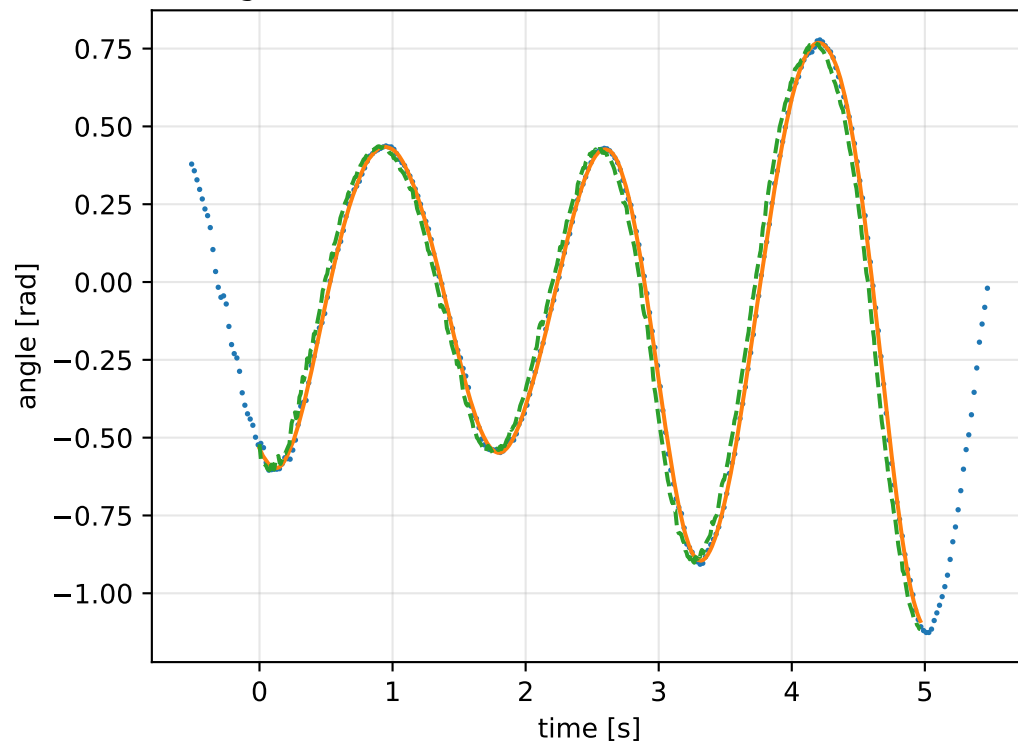
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

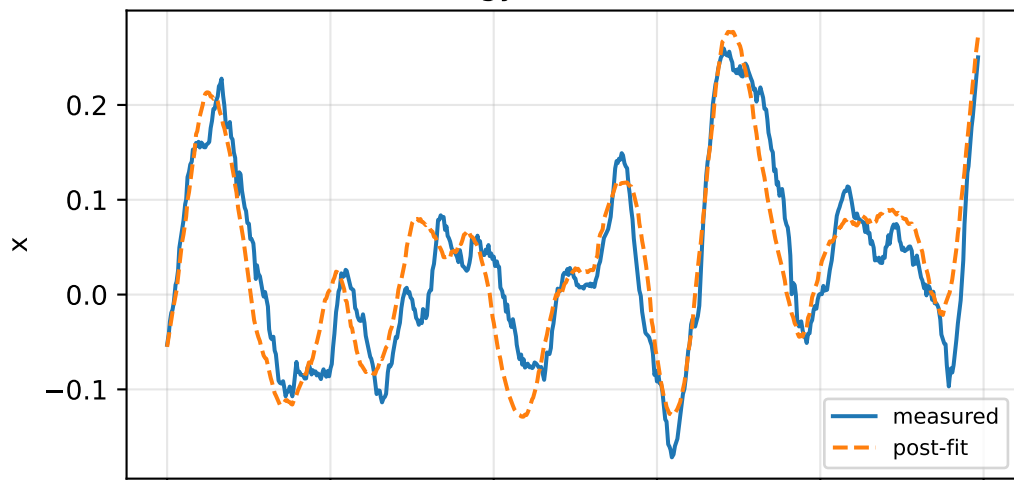


gimbal 4: RMSE=0.07131, max=0.1737 rad

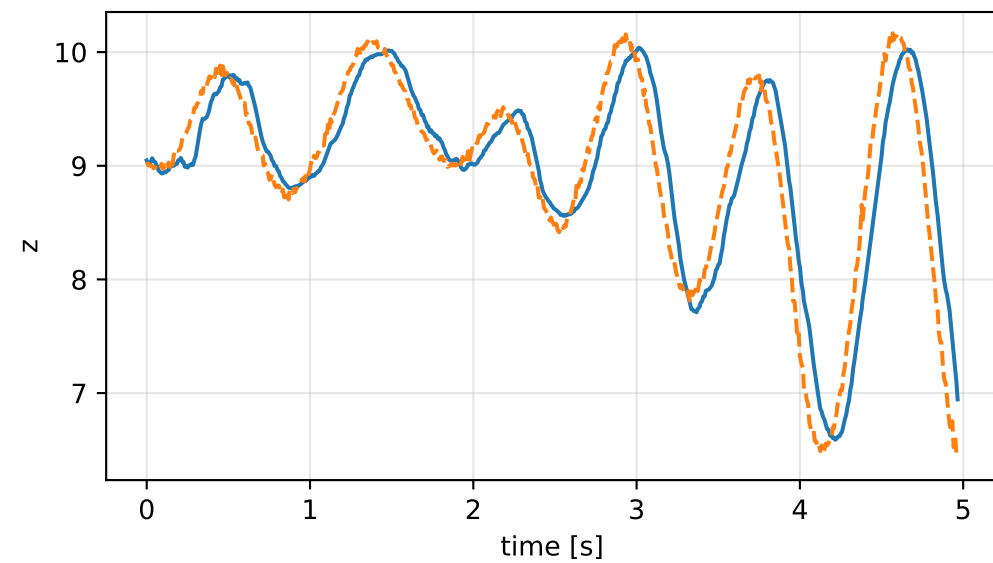
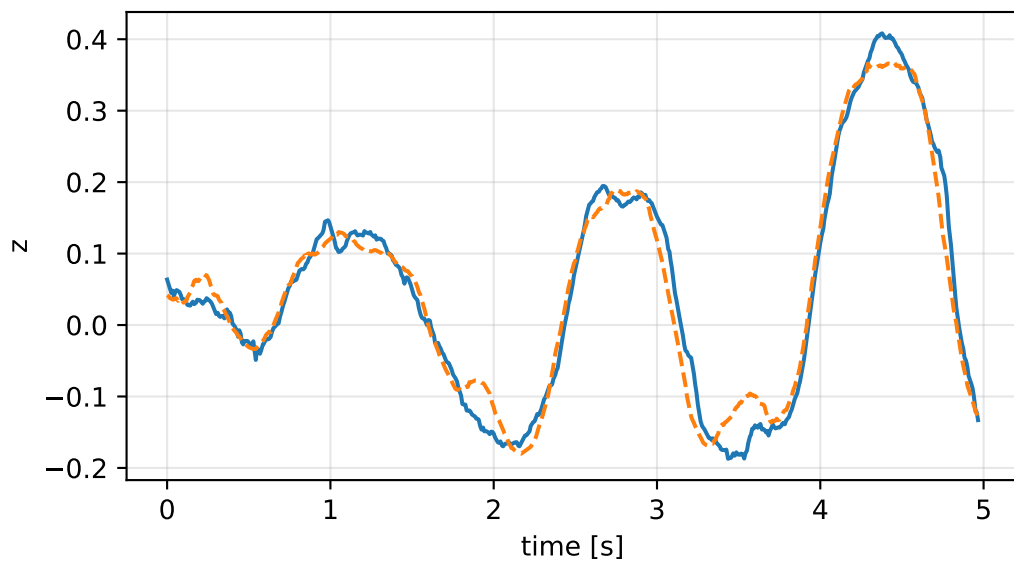
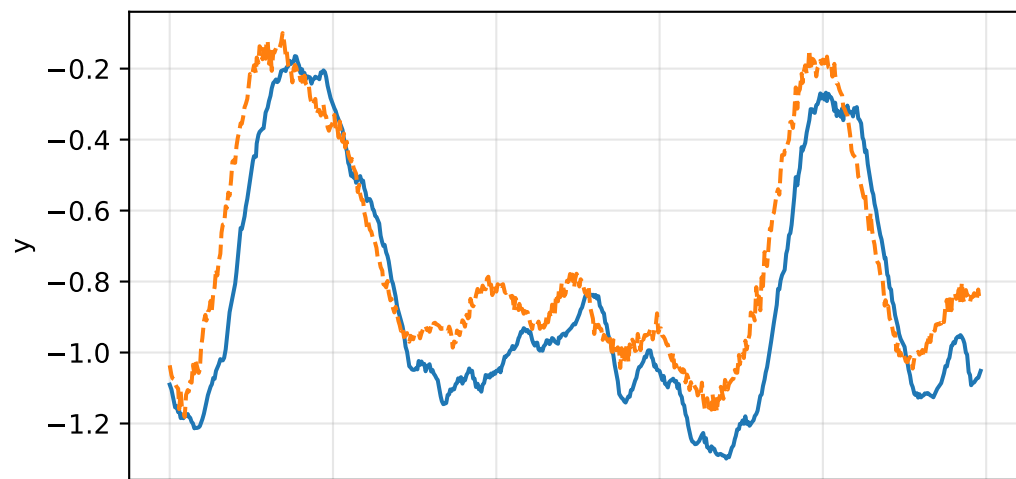
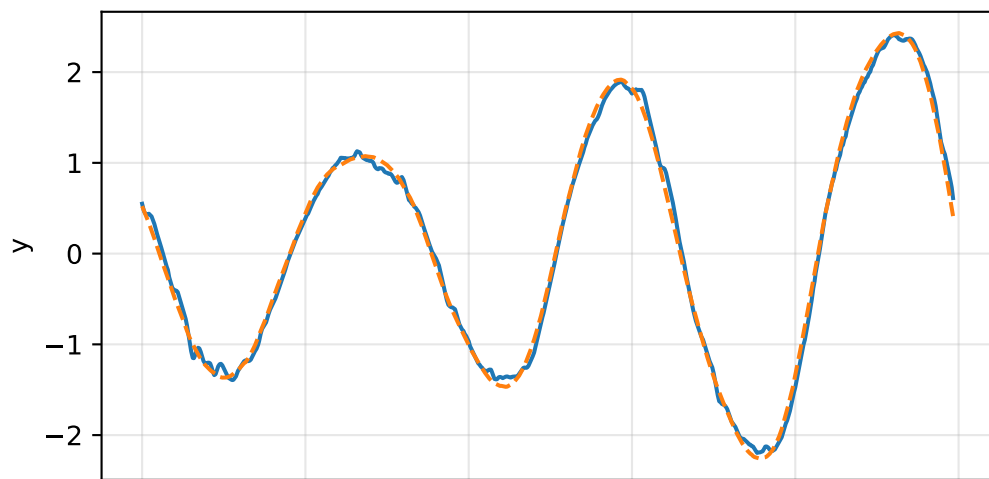
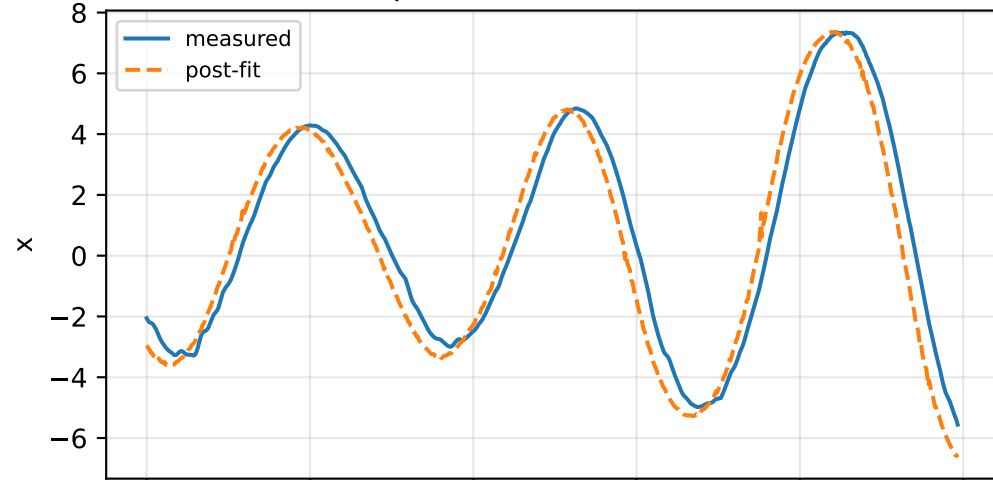


gimbal_measured_linear: IMU comparison (diagnostic only)

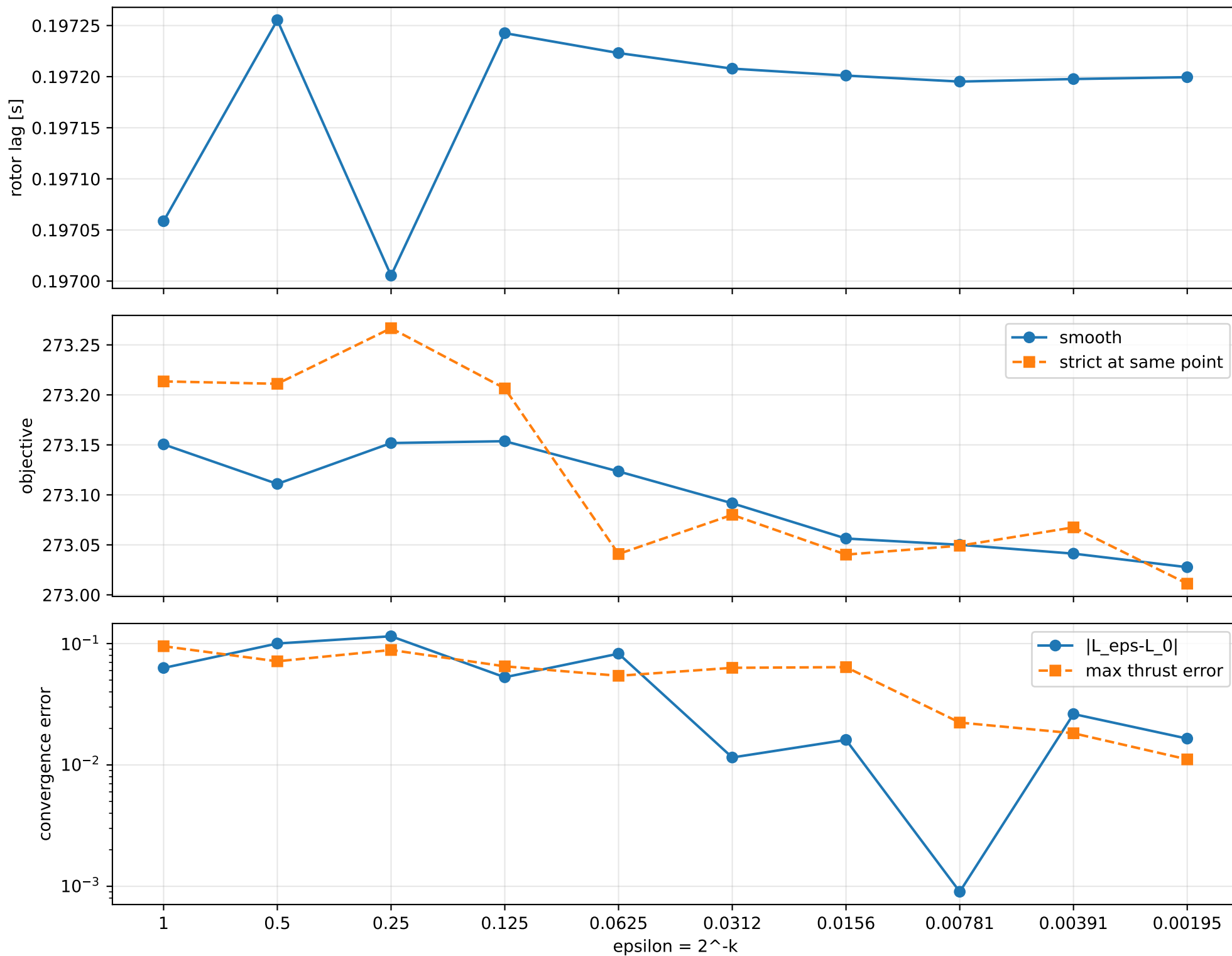
gyro [rad/s]



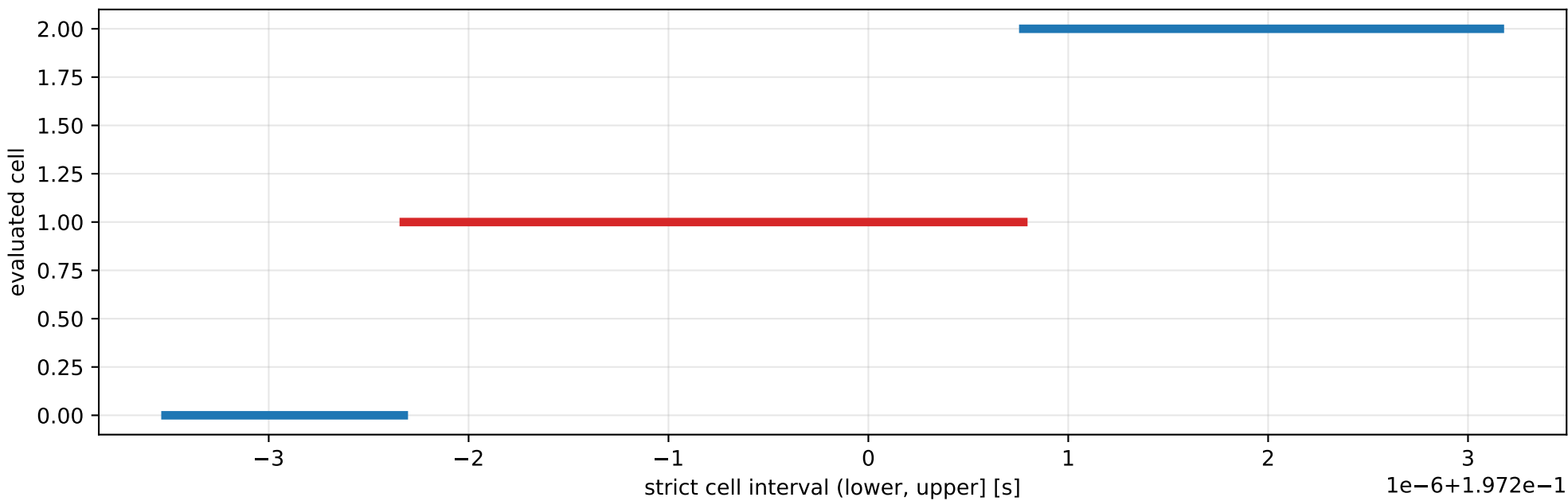
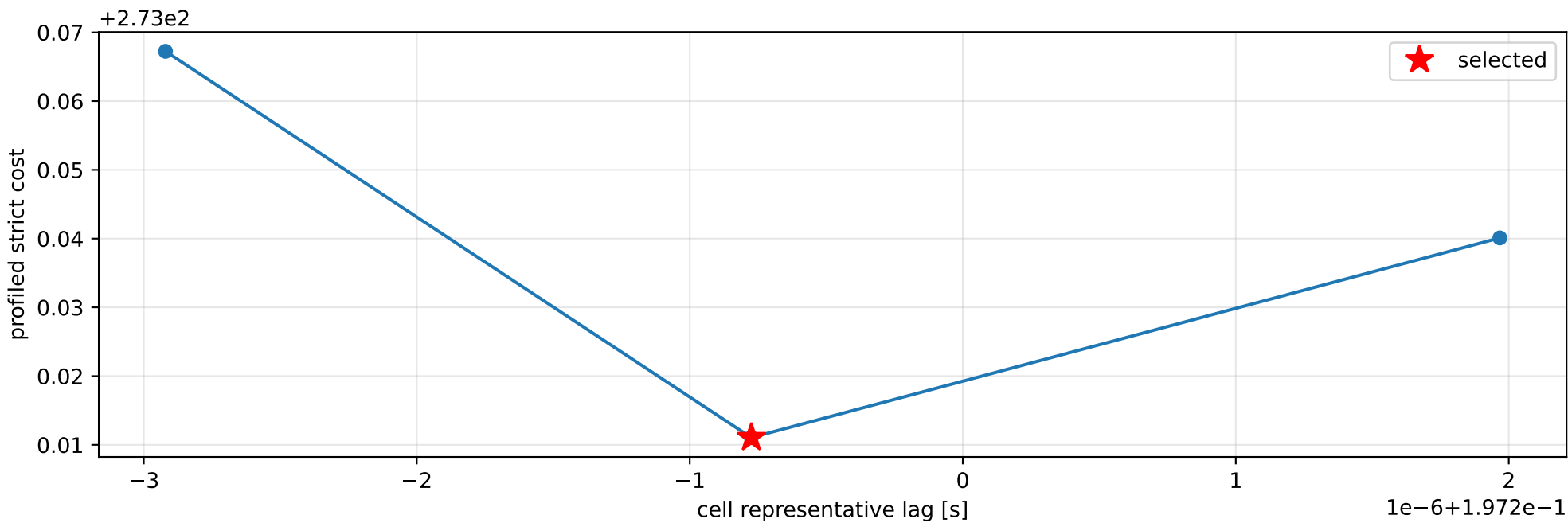
specific force [m/s^2]



gimbal_measured_linear: smooth-to-strict continuation



gimbal_measured_linear: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199485 s; data boundary=False

gimbal_measured_linear: parameter estimate

Case: gimbal_measured_linear

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.83203598

inertia [kg m²]:

```
[[ 0.49388125 -0.00161766 -0.07293049]
 [-0.00161766  0.37986163  0.00198012]
 [-0.07293049  0.00198012  0.84459821]]
```

CoG body [m]: [0.01810303 -0.00496539 -0.04562719]

force effectiveness: [1.62085446 2.09598703 1.81720943 1.49849357]

scale-free J/m [m²]:

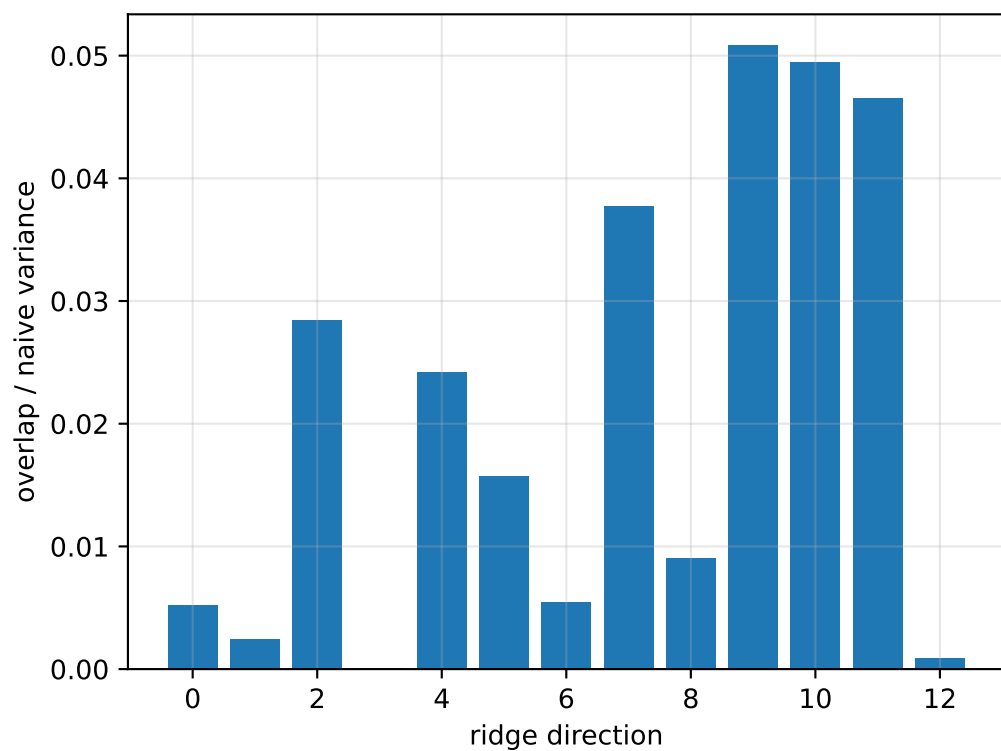
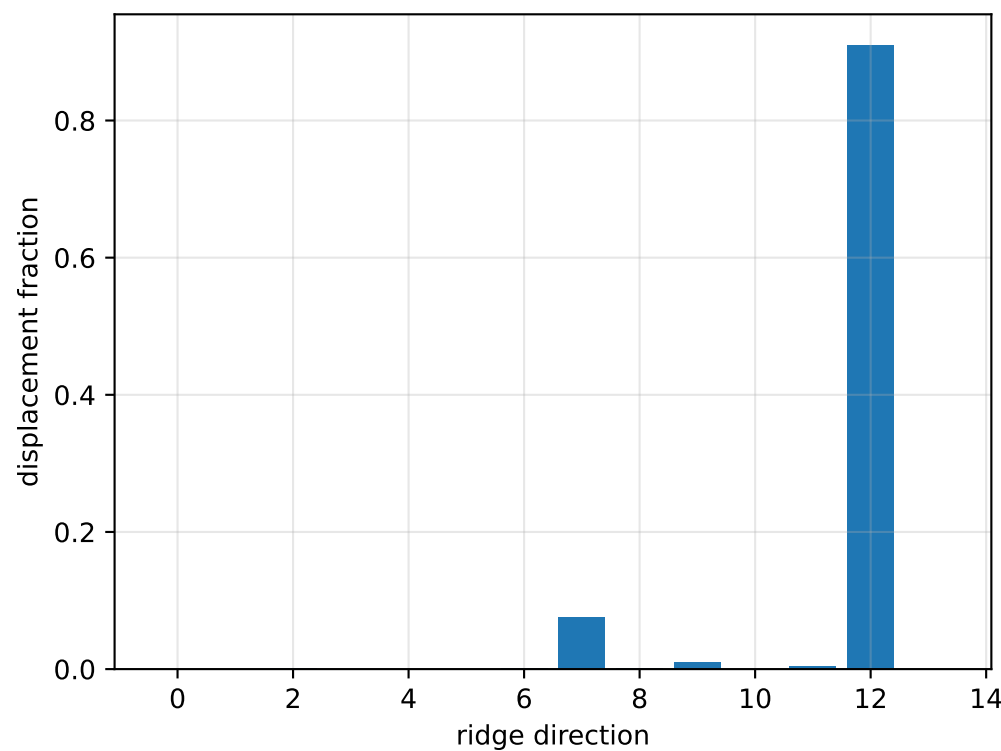
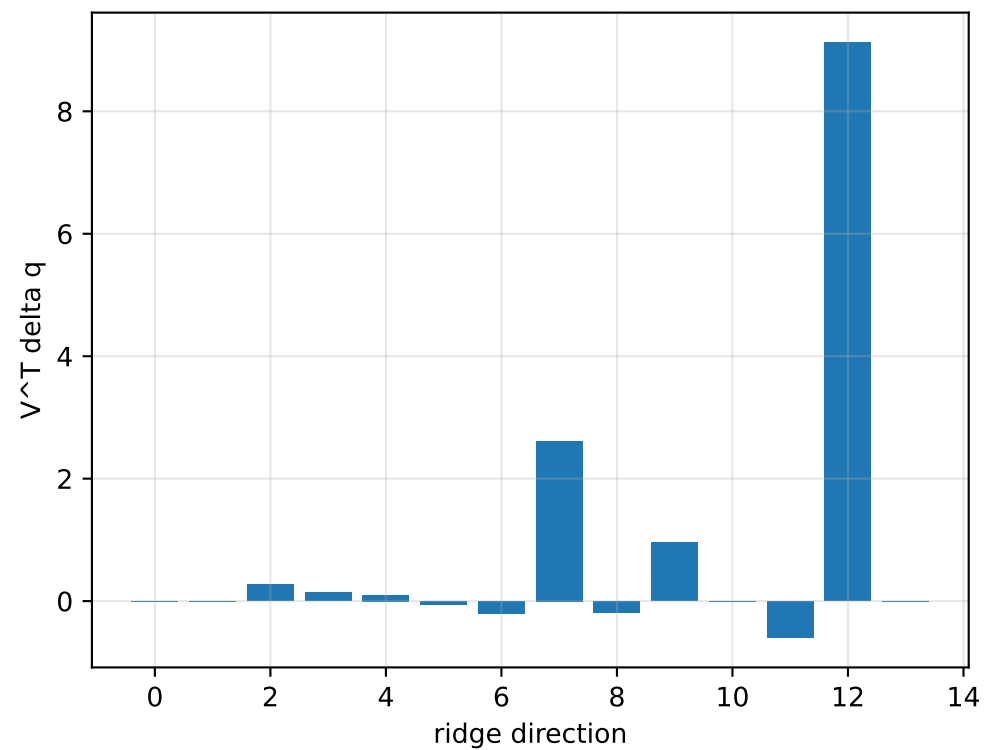
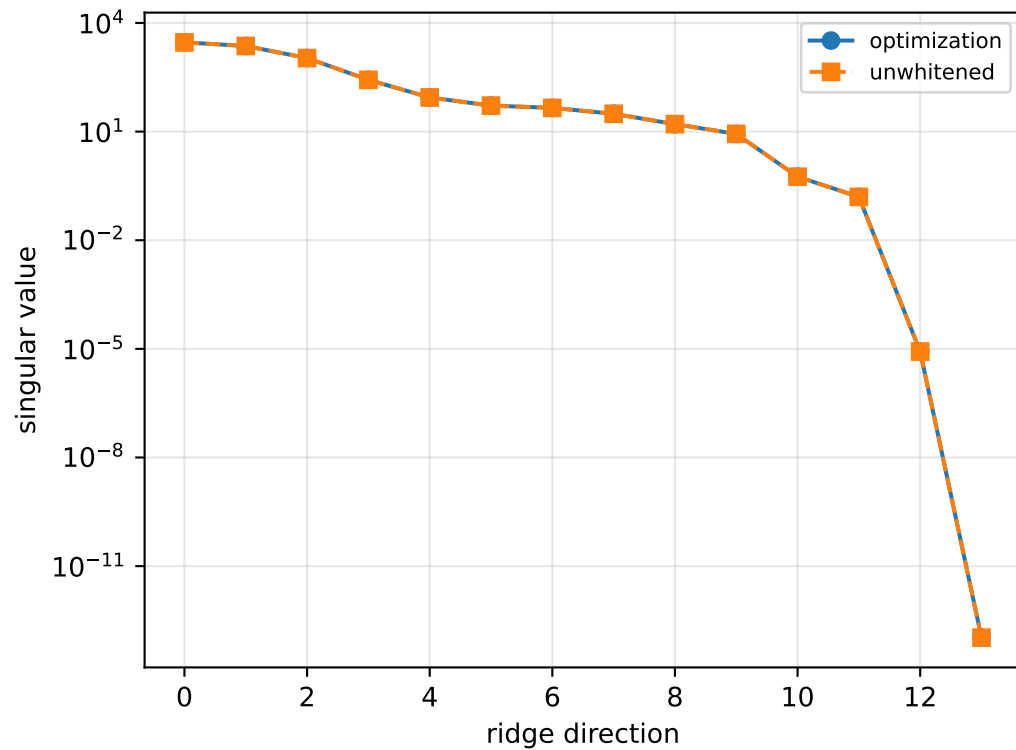
```
[[ 0.10220976 -0.00033478 -0.01509312]
 [-0.00033478  0.07861316  0.00040979]
 [-0.01509312  0.00040979  0.17479137]]
```

scale-free f/m: [0.33543924 0.43376892 0.37607531 0.31011639]

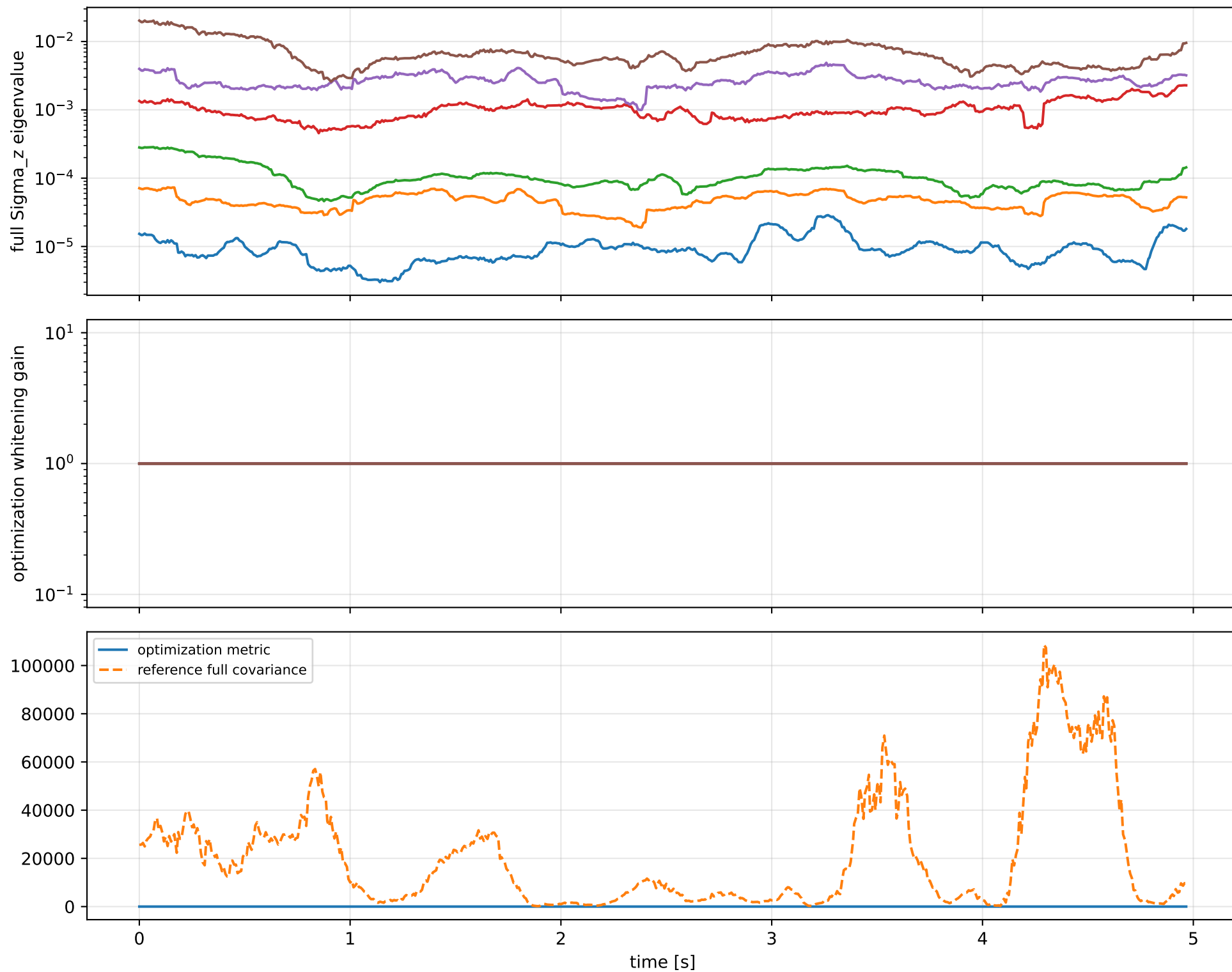
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

gimbal_measured_linear: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=8.7\text{e-}14$)



gimbal_measured_linear: optimization vs reference covariance



gimbal_measured_linear: wrench / closure (force max $2.09\text{e-}14$, torque max $8.71\text{e-}16$)

